

Start Here

Session 3 · Data Modeling & Warehouse Engineering

Welcome to the Vayu Air challenge

This pack is a hands-on data modeling challenge built around Vayu Air, an airline. You will turn raw source tables into a clean analytics warehouse: a star schema, a snowflaked geography, a slowly changing dimension, a partitioned fact, medallion layers, and a data contract. You will work in T-SQL on SQL Server.

What's in this pack

File	What it's for
00 · Start Here	This overview
01 · Problem Statement	The business background and what you're asked to do
02 · Assignment Questions	The 7 tasks to complete
03 · Importing the Dataset	Step-by-step load of the source data into SQL Server
04 · Data Dictionary	Every source table and column explained
05 · Hints	A nudge per question: ideas and patterns, never the answer
data/	The six source CSV files and a load.sql helper

How to work through it

1. Read the **Problem Statement** so the business context is clear.
2. Follow **Importing the Dataset** to load the bronze source tables.
3. Work through the **Assignment Questions** in order. They build a warehouse step by step.
4. Reach for the **Hints** only after you have tried a question yourself.

What to submit

1. Put **all your solution files** (your `.sql` scripts and a short results write-up) in a **single public GitHub repository**, and submit the **repository link**. That link is your submission.
2. For the partitioning task, include a **screenshot of the execution plan** showing the

partition counts for both queries.

3. *(Optional, good to have)* Add a short **video** (a Loom or any screen recording) walking through your model and your decisions. It is excellent practice for the stakeholder communication a data engineer does every day, and it strengthens your submission.

How your work is graded

Axis	What we look for
Correctness	Your model and queries meet the acceptance criteria against the data
Logic & approach	Right modeling choice: grain, surrogate keys, star vs snowflake, SCD2
Readability	Clear DDL, sensible names, comments, the grain stated up front
Communication	Your write-up or video explains your design to a stakeholder

Tip AI tools like Copilot, Claude, and Cursor are allowed and encouraged. What we assess is your understanding, so be ready to explain your grain choice, your SCD2 logic, and why partitioning helps.